Mobile Manipulation Project – Task Description and Evaluation Criteria

This document describes the set of tasks to be completed as part of the Mobile Manipulation Project. Each task evaluates different aspects of robot locomotion, motion planning, perception, and manipulation. Performance will be assessed based on functionality, accuracy, robustness, and time constraints.

Task 1 – Locomotion (Basic Functionality Test)

Environment

- No obstacles are present.

Given

- Two poses:
  - Point A: (x0 ,y0 ,α)
  - Point B: (x1 ,y1 ,β)

Objective

1. The robot must move from point A to point B.
2. The robot must then move from point B back to point A in reverse.

Variations

- Two different trials with different point pairs (A, B).

Evaluation

- **10 points** (functionality test)

Task 2 – Motion Planning with Obstacle Avoidance

Environment

- Obstacles are present, and their locations are known.

Given

- Three poses:
  - Point A: (x0 ,y0 ,α)
  - Point B: (x1 ,y1 ,β)

- Point C: (x2 ,y2 ,γ)

Objective

1. Navigate from point A to point B while avoiding obstacles.
2. Navigate from point B to point C while avoiding obstacles.

Variations

- Two different trials with different points A, B, C and different obstacle configurations.

Evaluation

- **5 points** (in-field challenge)

Task 3 – Motion Planning with Pick and Place

Environment

- Obstacles are present, and their locations are known.

Given

- Start and intermediate poses:
  - Point A: (x0 ,y0 ,α)
  - Point B: (x1 ,y1 ,β) — location of the object center
- Goal location:
  - Point C: (x2 ,y2) (orientation is free)
- Object location is known; object orientation is unknown.

Objective

1. Navigate from point A to point B while avoiding obstacles.
2. Pick up the object using the manipulator.
3. Move the object and place it at point C in any orientation.

Trials

- Two attempts are allowed if the first attempt fails.

Evaluation

- **5 points** (functionality test)

Task 4 – Navigation with Pick and Place

Environment

- Obstacles are present, and their locations are known.
- One object is present at an unknown location.

Given

- Start pose:
  - Point A: (x0 ,y0 ,α)
- Goal location:
  - Point C: (x2 ,y2) (orientation is free)

Objective

1. Navigate from point A to point B, where the object is located (unknown a priori).
2. Detect and pick the object.
3. Transport and place the object at point C.

Bonus

- Full score can be achieved if the object is placed at C with a fixed orientation, even if other minor errors occur.

Trials

- Two attempts are allowed if the first attempt fails.

Evaluation

- **5 points** (in-field challenge)

Task 5 – Object Detection, Navigation, and Pick and Place

Environment

- Three objects are present at unknown locations.
- Obstacles are present at known locations.

Given

- Start pose:
  - Point A: (x0 ,y0 ,α)
- Goal location:
  - Point C: (x2 ,y2) (orientation is free)

- The required order of object colors is given before the task begins.

Objective

1. Navigate from point A to the first object location (unknown).
2. Detect, pick, and place the object at point C.
3. Repeat the process for all three objects, following the specified color order.

Trials

- Two attempts are allowed if the first attempt fails.

Evaluation

- **10 points** (in-field challenge)
- **5 points** (functionality test)

Acceptable Errors and Scoring Penalties

1. Robot Localization Accuracy

- ≤ 10 cm error: **Full points**
- 10–20 cm error: **Half points**

2. Robot Orientation Accuracy

- ≤ 45° error: **Full points**
- 45°–90° error: **Half points**

3. Object Placement Accuracy

- ≤ 1.5 cm error: **Full points**
- ≤ 5 cm error: **Half points**

4. Obstacle Collisions

- For each obstacle collision, **1 point is deducted** from the corresponding task score.

5. Motion Smoothness

- If the robot stops or slows down considerably during task execution, **1 point is deducted**.

5. Bonus

- We will provide more information about the bonus after the last milestone

Time Constraints

- **Maximum allowed time for Tasks 1–4:** 3 minutes per task
- **Maximum allowed time for Task 5:** 10 minutes

Failure to complete a task within the allowed time results in a **5-point penalty**.